

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Mathematics — SSC-I (9th Grade)

Syllabus: Chapter 8 — Geometry of Straight Lines

ANSWER KEY (Version 1)

Total Marks: 65

Chapter 8

SECTION A — MCQ Answer Key (12 Marks)

#	Ans	Correct Option	Explanation
1	A	rise ÷ run	Gradient = vertical distance / horizontal distance = rise/run = $\tan\theta$. Option B inverts the fraction; C multiplies instead of divides; D adds them, which is meaningless.
2	C	0	A horizontal line has inclination $\theta = 0^\circ$, so gradient $m = \tan 0^\circ = 0$. A vertical line has undefined gradient. Distractors 1 and -1 belong to lines inclined at 45° and 135° respectively.
3	D	$m_1 \times m_2 = -1$	For perpendicular lines , $m_1 \times m_2 = -1$ (product of slopes is negative one). Equal slopes (A) means parallel. Sum zero (B) is incorrect. Product $+1$ (C) is a common misconception.
4	A	$y = mx + c$	The slope-intercept form is $y = mx + c$, where m is the slope and c is the y-intercept. Option B is point-slope form; C is two-intercept form; D is general form.
5	A	6/5	For $ax + by + c = 0$, slope = $-a/b$. Here $6x - 5y + 15 = 0$, so slope = $-(6)/(-5) = 6/5$. Option B gives the negative, confusing the sign; C and D invert the ratio.
6	B	$x = 3$	A line parallel to the y-axis has equation $x = \text{constant}$. Through $(3, 5)$ that constant is 3, giving $x = 3$. A line $y = 5$ would be parallel to the x-axis. Common error: confusing which coordinate gives the constant.
7	A	5	The x-intercept is found by setting $y = 0$: $x + 0 = 5$, so $x = 5$. Students often confuse x-intercept (set $y = 0$) with y-intercept (set $x = 0$), giving -5 or 1 as errors.
8	A	$(m_2 - m_1) / (1 + m_1 m_2)$	The angle from l_1 to l_2 uses $\tan\theta = (m_2 - m_1) / (1 + m_1 m_2)$. Option B has addition in numerator and subtraction in denominator (both wrong). Options C and D are incorrect forms.
9	B	Slope of AB = slope of BC	Three points are collinear if and only if slopes of any two segments joining them are equal. Option A (unequal slopes) means non-collinear. Option C mixes lengths with slopes. Option D means perpendicular.
10	C	Length of perpendicular from origin	In the normal form $x \cos\alpha + y \sin\alpha = p$, the value p is the length of the perpendicular from the origin to the line. It is always positive. Confusing it with slope or intercept is a common error.
11	B	x-axis	The equation $y = 0$ means every point on the line has y-coordinate zero, which is the definition of the x-axis. The y-axis is $x = 0$. This is directly from Miscellaneous Exercise 8 of the textbook.
12	A	(0, 0)	Both equations $9x - 7y = 0$ and $8x - 11y = 0$ pass through the origin (substituting $x = 0$, $y = 0$ satisfies both). Any two lines passing through the origin intersect there. Directly from Miscellaneous Exercise 8.

SECTION B — Short Question Answers (33 Marks)

2(i) 3 marks

Find the gradient of line whose inclination is 120° .

Gradient = $\tan \theta = \tan 120^\circ$ 1 mark

$\tan 120^\circ = \tan(180^\circ - 60^\circ) = -\tan 60^\circ = -\sqrt{3} \approx -1.732$ 1 mark

The gradient is negative because $90^\circ < 120^\circ < 180^\circ$, and $\tan$ is negative in the second quadrant. Lines making obtuse angles with the positive x-axis have negative gradients. 1 mark

OR

Find inclination of a line whose slope is -1.732 .

$m = \tan \theta = -1.732$ 1 mark

$\theta = \tan^{-1}(-1.732)$ 1 mark

Since $-1.732 \approx -\sqrt{3} = -\tan 60^\circ$, therefore **$\theta = 180^\circ - 60^\circ = 120^\circ$** 1 mark

2(ii) 3 marks

Find gradient and inclination of the line joining $A(2, 6)$ and $B(5, 8)$.

Gradient $m = (y_2 - y_1) / (x_2 - x_1) = (8 - 6) / (5 - 2) = 2/3$ 1 mark

$\tan \theta = 2/3 \Rightarrow \theta = \tan^{-1}(2/3)$ 1 mark

$\theta \approx 33.69^\circ$ 1 mark

OR

Find gradient and inclination of the line joining $C(-2, 4)$ and $D(1, -3)$.

$m = (-3 - 4) / (1 - (-2)) = -7/3$ 1 mark

$\tan \theta = -7/3 \Rightarrow \theta = \tan^{-1}(-7/3)$ 1 mark

Since value is negative, **$\theta \approx 180^\circ - 66.8^\circ = 113.2^\circ$** 1 mark

2(iii) 3 marks

$A(-2, 6)$, $B(7, -3)$: slopes of parallel and perpendicular lines.

Slope of $AB = (-3 - 6) / (7 - (-2)) = -9/9 = -1$ 1 mark

(a) Slope of line parallel to $AB = -1$ (parallel lines have equal slopes) 1 mark

(b) Slope of line perpendicular to AB : $m_1 \times m_2 = -1 \Rightarrow m_2 = -1/(-1) = 1$ 1 mark

OR

Find x if slope of line through $A(3, x)$ and $B(5, 8)$ is 4.

$m = (8 - x) / (5 - 3) = 4$ 1 mark

$(8 - x) / 2 = 4 \Rightarrow 8 - x = 8$ 1 mark

$x = 0$ 1 mark

2(iv) 3 marks

Prove $X(0, -3)$, $Y(4, 7)$, $Z(6, 12)$ are collinear using slopes.

Slope of $XY = (7 - (-3)) / (4 - 0) = 10/4 = 5/2$ 1 mark

Slope of $YZ = (12 - 7) / (6 - 4) = 5/2$ 1 mark

Since slope of $XY =$ slope of $YZ = 5/2$, the points **X, Y, Z are collinear.** 1 mark

OR

Find y if $P(4, y)$, $Q(5, 2)$, $R(6, 2y+1)$ are collinear.

Slope of $PQ = (2 - y) / (5 - 4) = 2 - y$ 1 mark

Slope of $QR = (2y+1 - 2) / (6 - 5) = 2y - 1$ 1 mark

For collinearity: $2 - y = 2y - 1 \Rightarrow 3 = 3y \Rightarrow y = 1$ 1 mark

2(v) 3 marks

Equation of line with slope 2 and y-intercept -3.

Using slope-intercept form: $y = mx + c$ **1 mark**

$$y = 2x + (-3) = y = 2x - 3 \quad \text{1 mark}$$

$$\text{General form: } 2x - y - 3 = 0 \quad \text{1 mark}$$

OR

Equation of line through $(-5, 7)$ with slope 4.

$$\text{Point-slope form: } y - y_1 = m(x - x_1) \quad \text{1 mark}$$

$$y - 7 = 4(x - (-5)) \Rightarrow y - 7 = 4(x + 5) \quad \text{1 mark}$$

$$y - 7 = 4x + 20 \Rightarrow 4x - y + 27 = 0 \quad \text{1 mark}$$

2(vi) 3 marks

Equation of line through $(2, 4)$ and $(-1, -4)$ using two-point form.

$$(y - 4) / (-4 - 4) = (x - 2) / (-1 - 2) \Rightarrow (y - 4) / (-8) = (x - 2) / (-3) \quad \text{1 mark}$$

$$-3(y - 4) = -8(x - 2) \Rightarrow -3y + 12 = -8x + 16 \quad \text{1 mark}$$

$$8x - 3y - 4 = 0 \quad \text{1 mark}$$

OR

Equation with x-intercept = -6 and y-intercept = 5.

$$\text{Two-intercept form: } x/a + y/b = 1 \Rightarrow x/(-6) + y/5 = 1 \quad \text{1 mark}$$

$$\text{Multiply both sides by } -30: 5x - 6y = -30 \quad \text{1 mark}$$

$$5x - 6y + 30 = 0 \quad \text{1 mark}$$

2(vii) 3 marks

Reduce $6x - 5y + 15 = 0$ to slope-intercept form.

$$6x - 5y + 15 = 0 \Rightarrow 5y = 6x + 15 \quad \text{1 mark}$$

$$y = (6/5)x + 3 \quad \text{1 mark}$$

$$\text{Slope } m = 6/5 \text{ and y-intercept } c = 3 \quad \text{1 mark}$$

OR

Reduce $4x - 3y + 9 = 0$ to two-intercept form.

$$4x - 3y = -9 \quad \text{1 mark}$$

$$(4x)/(-9) + (-3y)/(-9) = 1 \Rightarrow x/(-9/4) + y/3 = 1 \quad \text{1 mark}$$

$$\text{x-intercept} = -9/4 \text{ and y-intercept} = 3 \quad \text{1 mark}$$

2(viii) 3 marks

Equation of line through $(3, -2)$, perpendicular to slope 3.

$$\text{Given slope } m_1 = 3; \text{ perpendicular slope } m_2 = -1/3 \quad \text{1 mark}$$

$$y - (-2) = -(1/3)(x - 3) \Rightarrow 3(y + 2) = -(x - 3) \quad \text{1 mark}$$

$$3y + 6 = -x + 3 \Rightarrow x + 3y + 3 = 0 \quad \text{1 mark}$$

OR

Equation of line with y-intercept = -7, parallel to slope 4.

$$\text{Parallel lines have equal slopes, so } m = 4 \quad \text{1 mark}$$

$$\text{Slope-intercept form: } y = 4x + (-7) \quad \text{1 mark}$$

$$y = 4x - 7 \text{ (or } 4x - y - 7 = 0) \quad \text{1 mark}$$

2(ix) 3 marks

Angle from l_1 to l_2 if $m_1 = -1$ and $m_2 = 2$.

$$\tan\theta = (m_2 - m_1) / (1 + m_1m_2) \quad \text{1 mark}$$

$$\tan\theta = (2 - (-1)) / (1 + (2)(-1)) = 3/(-1) = -3 \quad \text{1 mark}$$

$$\theta = \tan^{-1}(-3) \Rightarrow \theta \approx 108.43^\circ \quad \text{1 mark}$$

OR

Angle between $3x + 2y + 5 = 0$ and $2x - 3y + 8 = 0$.

$$m_1 = -3/2 \text{ (from first line); } m_2 = 2/3 \text{ (from second line)} \quad \text{1 mark}$$

$$m_1 \times m_2 = (-3/2)(2/3) = -1 \quad \text{1 mark}$$

Since $m_1m_2 = -1$, the lines are perpendicular: **angle = 90°** 1 mark

2(x) 3 marks

Point of intersection: $2x + y + 1 = 0$ and $x - y - 4 = 0$.

$$\text{Adding the two equations: } 3x - 3 = 0 \Rightarrow x = 1 \quad \text{1 mark}$$

$$\text{Substituting } x = 1 \text{ into equation 2: } 1 - y - 4 = 0 \Rightarrow y = -3 \quad \text{1 mark}$$

Point of intersection = (1, -3) 1 mark

OR

Point of intersection: $x + y + 3 = 0$ and $2x - 5y + 8 = 0$.

$$\text{From eq.1: } x = -y - 3. \text{ Substituting into eq.2: } 2(-y - 3) - 5y + 8 = 0 \quad \text{1 mark}$$

$$-2y - 6 - 5y + 8 = 0 \Rightarrow -7y + 2 = 0 \Rightarrow y = 2/7 \quad \text{1 mark}$$

$$x = -(2/7) - 3 = -23/7 \Rightarrow \text{Point} = (-23/7, 2/7) \quad \text{1 mark}$$

2(xi) 3 marks

Line through intersection of $x + 2y = 0$ and $x - y = 3$, passing through (1, 1).

$$\text{Family of lines: } (x + 2y) + k(x - y - 3) = 0 \quad \text{1 mark}$$

$$\text{Substituting (1, 1): } 3 + k(1 - 1 - 3) = 0 \Rightarrow 3 - 3k = 0 \Rightarrow k = 1 \quad \text{1 mark}$$

$$(x + 2y) + (x - y - 3) = 0 \Rightarrow \mathbf{2x + y - 3 = 0} \quad \text{1 mark}$$

OR

Family through intersection of $6x + 5y + 3 = 0$ and $2x - 5y + 13 = 0$ with slope 3.

$$\text{Family: } (6x + 5y + 3) + k(2x - 5y + 13) = 0 \Rightarrow (6+2k)x + (5-5k)y + (3+13k) = 0 \quad \text{1 mark}$$

$$\text{Slope} = -(6+2k)/(5-5k) = 3 \Rightarrow -6 - 2k = 15 - 15k \Rightarrow 13k = 21 \Rightarrow k = 21/13 \quad \text{1 mark}$$

$$\text{Substituting } k: 9x + 12y - 6 = 0 \Rightarrow \mathbf{3x + 4y - 2 = 0} \quad \text{1 mark}$$

SECTION C — Long Question Answers (20 Marks)

3(i) 5 marks

$A(3, -1), B(-5, -5), C(1, 3)$: prove right-angled triangle, identify the right angle.

Step 1: Find slopes of all three sides 2 marks

$$\text{Slope of AB} = (-5 - (-1)) / (-5 - 3) = -4 / -8 = 1/2$$

$$\text{Slope of BC} = (3 - (-5)) / (1 - (-5)) = 8 / 6 = 4/3$$

$$\text{Slope of AC} = (3 - (-1)) / (1 - 3) = 4 / -2 = -2$$

Step 2: Check perpendicularity 2 marks

$$m(\text{AB}) \times m(\text{AC}) = (1/2) \times (-2) = -1 \quad \checkmark \quad (\text{perpendicular})$$

$$m(\text{BC}) \times m(\text{AB}) = (4/3) \times (1/2) = 2/3 \neq -1$$

$$m(\text{BC}) \times m(\text{AC}) = (4/3) \times (-2) = -8/3 \neq -1$$

Conclusion: Since $\text{slope}(\text{AB}) \times \text{slope}(\text{AC}) = -1$, lines AB and AC are perpendicular. The **right angle is at vertex A(3, -1)**. **1 mark**

OR

$A(-2, 1), B(6, 3), C(10, 5), D(2, 3)$: prove parallelogram.

$$\text{Slope of AB} = (3-1)/(6-(-2)) = 2/8 = 1/4$$

$$\text{Slope of DC} = (5-3)/(10-2) = 2/8 = 1/4 \Rightarrow \text{AB} \parallel \text{DC} \quad \checkmark$$

$$\text{Slope of BC} = (5-3)/(10-6) = 2/4 = 1/2$$

$$\text{Slope of AD} = (3-1)/(2-(-2)) = 2/4 = 1/2 \Rightarrow \text{BC} \parallel \text{AD} \quad \checkmark$$

Both pairs of opposite sides are parallel. **ABCD is a parallelogram.** **5 marks total**

3(ii) 5 marks

Reduce $6x - 5y + 15 = 0$ into all six standard forms.

(i) Slope-intercept form 1 mark

$5y = 6x + 15 \Rightarrow y = (6/5)x + 3$; slope = $6/5$, y-intercept = 3 **(ii) Point-slope form** (using point (0, 3))

$y - 3 = (6/5)(x - 0)$ **(iii) Two-point form** (using (0, 3) and $(-5/2, 0)$)

$(y - 3)/(0 - 3) = (x - 0)/(-5/2 - 0)$ **1 mark** **(iv) Two-intercept form**

$6x - 5y = -15 \Rightarrow x/(-5/2) + y/3 = 1$ **1 mark** **(v) Normal form**

Rewrite as $-6x + 5y = 15$. Divide by $\sqrt{(36+25)} = \sqrt{61}$:

$x(-6/\sqrt{61}) + y(5/\sqrt{61}) = 15/\sqrt{61}$, where $p = 15/\sqrt{61}$ **1 mark** **(vi) Symmetric form** (using point (0, 3),

$\sin\theta = 6/\sqrt{61}$, $\cos\theta = 5/\sqrt{61}$)

$(x - 0)/(5/\sqrt{61}) = (y - 3)/(6/\sqrt{61})$ **1 mark**

OR

Perpendicular bisector of segment joining $A(0, 6)$ and $B(2, -2)$.

$$\text{Midpoint M} = ((0+2)/2, (6+(-2))/2) = (1, 2)$$

$$\text{Slope of AB} = (-2-6)/(2-0) = -8/2 = -4$$

$$\text{Slope of perpendicular bisector} = -1/(-4) = 1/4$$

Equation: $y - 2 = (1/4)(x - 1) \Rightarrow 4y - 8 = x - 1 \Rightarrow x - 4y + 7 = 0$

x-intercept (set $y = 0$): $x = -7 \Rightarrow$ meets x-axis at **(-7, 0)**

y-intercept (set $x = 0$): $-4y = -7 \Rightarrow y = 7/4 \Rightarrow$ meets y-axis at **(0, 7/4)** **5 marks total**

3(iii) 5 marks

Interior angles of triangle ABC: $A(-2, 0)$, $B(3, 0)$, $C(6, 5)$.

Step 1: Find slopes of all sides 1 mark

$$m_1 = \text{slope of AB} = (0-0)/(3-(-2)) = 0/5 = 0$$

$$m_2 = \text{slope of BC} = (5-0)/(6-3) = 5/3$$

$$m_3 = \text{slope of AC} = (5-0)/(6-(-2)) = 5/8$$

Step 2: Angle at A (from AB to AC) 1 mark

$$\tan \alpha = (m_3 - m_1) / (1 + m_1 \cdot m_3) = (5/8 - 0) / (1 + 0) = 5/8$$

$$\alpha = \tan^{-1}(5/8) \approx 32^\circ$$

Step 3: Angle at B (from BA to BC; $m(\text{BA}) = 0$, $m(\text{BC}) = 5/3$) 1 mark

$$\tan \beta = (m_2 - m_1) / (1 + m_1 \cdot m_2) = (5/3 - 0) / (1 + 0 \cdot 5/3) = 5/3$$

$$\beta = \tan^{-1}(5/3) \approx 59^\circ$$

Step 4: Angle at C (by subtraction) 1 mark

$$\gamma = 180^\circ - 32^\circ - 59^\circ = 89^\circ \approx 89^\circ$$

Verification: $32^\circ + 59^\circ + 89^\circ = 180^\circ \checkmark$ 1 mark

OR

Interior angles of triangle XYZ: $X(-2, 3)$, $Y(-3, -4)$, $Z(5, 2)$.

Step 1: Find slopes

$$\text{Slope of XY} = (-4-3)/(-3-(-2)) = -7/-1 = 7$$

$$\text{Slope of YZ} = (2-(-4))/(5-(-3)) = 6/8 = 3/4$$

$$\text{Slope of XZ} = (2-3)/(5-(-2)) = -1/7$$

Step 2: Check for right angle at X

$$m(\text{XY}) \times m(\text{XZ}) = 7 \times (-1/7) = -1 \quad \checkmark \rightarrow \text{XY} \perp \text{XZ} \rightarrow \text{angle at X} = 90^\circ$$

Step 3: Angle at Y (from YX to YZ; $m(\text{YX}) = 7$, $m(\text{YZ}) = 3/4$)

$$\tan(Y) = (m_{\text{YZ}} - m_{\text{YX}}) / (1 + m_{\text{YX}} \cdot m_{\text{YZ}})$$

$$= (3/4 - 7) / (1 + 7 \cdot 3/4)$$

$$= (-25/4) / (25/4) = -1$$

$$\text{angle at Y} = \tan^{-1}(-1) \rightarrow \text{since obtuse context: } Y = 180^\circ - 45^\circ = 135^\circ$$

Step 4: Angle at Z

$$\gamma = 180^\circ - 90^\circ - 135^\circ = -45^\circ \rightarrow \text{the formula gave the reflex supplement;}$$

$$\text{use the acute value directly: angle at Z} = 45^\circ$$

$$\text{Verification: } 90^\circ + 135^\circ + 45^\circ = 270^\circ \rightarrow Y \text{ is measured as the exterior supplement.}$$

$$\text{Interior angles: } X = 90^\circ, Y = 45^\circ, Z = 45^\circ \rightarrow \text{Sum} = 180^\circ \checkmark$$

$$\text{Note: the formula } \tan \theta = (m_2 - m_1)/(1 + m_1 m_2) \text{ measures the angle from } l_1 \text{ to } l_2$$

$$\text{in anti-clockwise direction, which can be obtuse. The interior angle at Y} = 180^\circ - 135^\circ = 45^\circ.$$

Interior angles: X = 90°, Y = 45°, Z = 45° (isosceles right triangle) 5 marks total

3(iv) 5 marks

Cricket team: 96 runs in 16 overs; 180 runs in 30 overs.

(i) Equation using two-point form 1 mark

Let x = overs, y = runs. Points: (16, 96) and (30, 180).

$$\text{Slope } m = (180 - 96)/(30 - 16) = 84/14 = 6$$

$$y - 96 = 6(x - 16) \rightarrow y - 96 = 6x - 96 \rightarrow y = 6x$$

(ii) Meaning of gradient: The slope 6 means the team scores **6 runs per over** on average. 1 mark **(iii)**

Predicted score after 45 overs 1 mark

$$y = 6 \times 45 = 270 \text{ runs}$$

(iv) Overs for 240 runs 1 mark

$$240 = 6x \rightarrow x = 40 \text{ overs}$$

Summary: $y = 6x$, gradient = 6 runs/over, score at 45 overs = 270, overs for 240 runs = 40.

1 mark

OR

Nasir: fruiters Rs. 120/dozen, Shakri Malta Rs. 150/dozen, earned Rs. 1200.

(i) Standard form equation

Let x = dozens of fruiters, y = dozens of Shakri Malta.

$$120x + 150y = 1200 \text{ (divide by 30)} \Rightarrow 4x + 5y = 40 \text{ 2 marks (ii) If } y = 4 \text{ dozen Shakri Malta}$$

$$4x + 5(4) = 40 \Rightarrow 4x = 20 \Rightarrow x = 5 \text{ dozens of fruiters 2 marks (iii) Slope-intercept form}$$

$4x + 5y = 40 \Rightarrow y = -(4/5)x + 8$. Slope = $-4/5$ means for each additional dozen of fruiters sold, Shakri Malta dozens decrease by $4/5$ (fixed budget trade-off). 1 mark